Individual Reflection

I completed both the Data Engineer and ML Engineer roles. I cleaned and split the transaction data, built a sparse purchase matrix, compared SVD with PCA, evaluated recommendations against best sellers, and created four exploratory customer groups.

I learned how sparse storage reduces memory usage and how PCA's centering differs from SVD. I also learned that reconstruction quality does not directly determine recommendation quality. SVD achieved a 22.30% hit rate at 10, compared with 17.38% for best sellers, while PCA performed almost identically. Holding out future purchases helped prevent data leakage.

I would improve support for new customers and rare products, evaluate additional time periods, and investigate segmentation beyond two components. Purchasing activity strongly influenced the current groups, so their marketing usefulness needs further validation.